SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105

Student Name: V.M.HARIHARA PRASATH

Reg. No.:192421393

Course Code: ITA0302 **Slot:D**

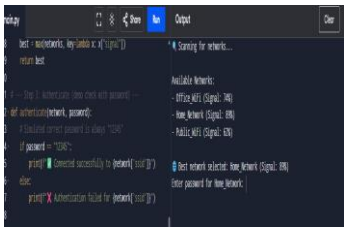
Course Name: MOBILE COMPUTING FOR MODERN NETWORKIN3G SYSTEM

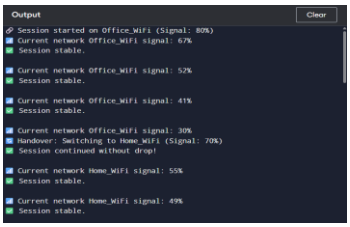
Course Faculty: Dr.SARAVANAN & Dr.RAMESHKUMAR

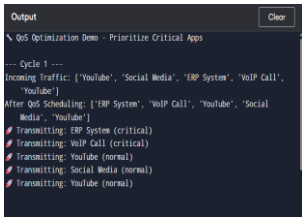
Project Title: A UNIFIED FRAMEWORK FOR SEAMLESS ROAMING AND CONNECTIVITY IN MOBILE COMPUTING

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

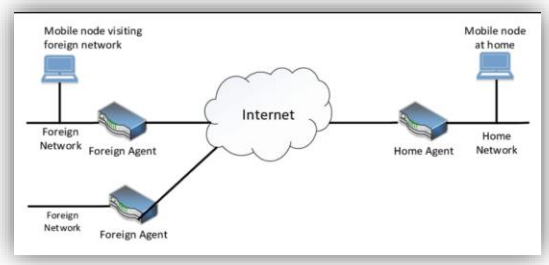
MODULE PHOTO





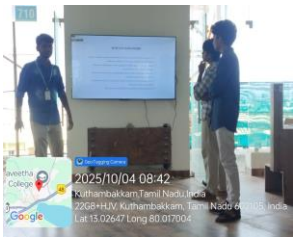


INDIVIDUAL STUDENT CONTRIBUTION MODULE WORK IN THE PROJECT



PRESENTATION IMAGE

REVIEW I



REVIEW II

Project Description: **SEAMLESS HANDOVER → SWITCH NETWORKS WITHOUT SESSION DROP**

The Module **"Seamless Handover for Uninterrupted Network Connectivity"** focuses on enabling users to **switch between networks without session drop**, ensuring continuous service availability. Using **advanced mobility management protocols** and **intelligent handover mechanisms**, the system maintains active sessions during transitions between Wi-Fi, 4G, and 5G networks. It leverages **real-time monitoring** of network conditions to select the best available connection dynamically, providing **low-latency and high-reliability communication**. The solution is designed to work efficiently in **high-mobility environments** such as vehicles, public transport, and remote areas, were connectivity changes frequently. The system incorporates **context-aware decision-making models** that consider factors like signal strength, bandwidth, and user activity to ensure **zero packet loss** and **minimal handover delay**. It integrates with **cloud-based network controllers** for intelligent routing and supports **multi-access edge computing (MEC)** to reduce latency during transitions.

Student Signature **Guide Signature**